

(NICU) and Special Care Nursery (SCN) from 1 January 2009 to 31 December 2013 was performed. Qualitative interviews were undertaken with parents of Aboriginal and Torres Strait Islander infants admitted to the unit from June 2014 to June 2015.

Results: The most common reason for admission of infants was prematurity (69.4%). While the majority of infants (86.1%) were born in Canberra, more than half (55.6%) resided outside Canberra. Four major themes emerged from the family interviews: the largely positive impact of the unit's physical environment; the complexity of communication needs; varied awareness and engagement with support services; and that having an infant in a neonatal unit is an inherently stressful situation.

Conclusion: Important steps have been made towards meeting the needs of Aboriginal and Torres Strait Islander families in the unit. It is necessary to continue to improve communication, increase awareness of support services and reduce the levels of stress experienced by these families.

PARENT CONCERNS FOR CHILD DEVELOPMENT AMONGST PRETERM GRADUATES OF NEONATAL CARE IN AUSTRALIA.

Megan Bater^{1*}, Michael Stark², Chad Andersen³, Nicolette Hodyl⁴

¹Robinson Research Institute / WCH Adelaide, the ²Robinson Research Institute, the ³Department of Neonatal Medicine, Women's and Children's Hospital, Adelaide, the ⁴Women's and Children's Hospital, Robinson Research Institute

Background: Having a preterm baby in the Neonatal Unit is an incredibly anxiety provoking time for parents. Concerns often continue following discharge to home, with many parents wondering whether their child has evaded morbidity. This research aimed to investigate parents' current concerns regarding the development of their child/ren, all graduates of neonatal care.

Method: Parents of children admitted to Australian neonatal intensive or special care completed an online questionnaire via Survey Monkey through the Miracle Babies Facebook page.

Results: Parents provided data on 484 child graduates of neonatal care. The mean gestational age was 30 weeks, mean current age 3 years (1 month–15 years). Parents expressed current concerns for the development of 191 (39.5%) children, which varied according to the child's current age and gestation. 24% of children had concerns reported in multiple domains. The most prevalent concerns related to developmental milestones (46%), motor (28%), speech/language (35%), diagnosis (28%), behaviour (25%), growth/feeding (16%) and that the issue was ongoing (18%). Parental exposure to more than one child receiving neonatal care did not increase parent likelihood of reporting a developmental concern, with no differences seen between this group and first time parents.

Conclusions: The current focus upon vision, hearing, cerebral palsy and respiratory morbidity in children born preterm does not fully address the key developmental concerns reported by parents of neonatal care graduates. Further educational resources to increase parents' knowledge of child development may prove helpful to alleviate parental anxiety, improve parenting confidence and optimise neurodevelopmental outcomes post discharge from neonatal care.

ABSENT DUCTUS VENOSUS: A CASE SERIES WITH NEONATAL OUTCOMES

Rhys Belling^{1*}, John Newnham¹, Chhaya Mehrotra¹

¹King Edward Memorial Hospital

Background: The ductus venosus delivers oxygenated blood from umbilical vein (UV) to fetus by shunting away from the hepatic circulation to the inferior vena cava. An absent ductus venosus (ADV) varies in severity, relating to the UA's altered course, travelling either intrahepatically or extrahepatically. Extrahepatic, bypassing the liver entirely, can include direct communication with atria or coronary sinus, and is more often associated with a poorer prognosis, due to this high volume flow directly into the heart¹.

Methods: A retrospective case series for all cases of ADV from database review of 5 years of ultrasound assessment at our institution, with case note review for neonatal outcomes.

Results: 5 cases were identified, all extrahepatic courses. 2 had severe neonatal outcomes; both had genetic abnormalities. One had a UA connecting to the coronary sinus, amniocentesis showing a mosaic trisomy 7 prompting 26 week termination, post-mortem showing significant cardiomegaly with dilated right atria, right ventricle, and coronary sinus. The other had a UA directly communicating with the right iliac vein, and post-natal diagnosis of Noonan Syndrome in fact accounts for associated pathologies identified, being bilateral pleural effusions, polyhydramnios, and pronounced hydrops which prompted caesarian section at 33 weeks. The remaining cases yielded good outcomes with brief admissions to nursery.

Conclusion: Extrahepatic connections to the right atria, or iliac veins, with associated polyhydramnios and hydrops constitute a poor prognosis however in this series constitutes an association rather than a causation for poor outcomes.

Reference:

1. Kiserud T, Acharya G. The fetal circulation. Prenatal diagnosis 2004; 24(13), 1049–59

THE SPINY MOUSE: A MODEL FOR MENSTRUATION AND REPRODUCTIVE AGEING

Nadia Bellofiore^{1*}, Jemma Evans¹, Peter Temple-Smith², Hayley Dickinson³

¹Hudson Institute of Medical Research, the ²Monash University, the ³Ritchie Centre

Background: Shedding of the superficial endometrium is a rare reproductive strategy, shared only by a handful of mammals, and most of these are higher order primates. Consequently, long term studies using animal models on menstrual cycle changes, particularly in regards to age and menopause, are lacking. We recently discovered the spiny mouse is the only rodent in existence to undergo menstruation. Hence, our aim is to determine the reproductive capacity of females across their lifespan.

Method: We subjected female spiny mice (n = 5–14 per group) of ages 4–6, 12, 24 and 36 months to daily vaginal lavage and weighing for 14 days to map physiological differences in the menstrual cycle. We compared average ovarian and total reproductive tract weights across all groups. We then collected these tissues for histological analysis.

Results: We observed significant increases in body weight in all ages compared to 4–6 month old females ($p < 0.0001$). The proportion of females who exhibit regular cycling are maintained at 75% for up to 12 months of age, declining to 57% at 24 months and 63% at 36 months. We found no differences in the cycle length, cycle stages or correlations with weight and cycle length between these ages. Morphological data on reproductive tissues and ovarian follicular quantification are ongoing.

Conclusions: Our preliminary results suggest the average age of reproductive decline in spiny mice is over 3 years. We will correlate ovarian reserve with age to determine whether the spiny mouse could serve as a pre-clinical model for menopause and to closely examine effects of age on fertility.

FETAL INTESTINAL VOLVULUS WITH MECONIUM PERITONITIS: A CASE REPORT

Emanuelle Best^{1*}, Cecelia O'Brien², Wendy Carseldine¹, Aniruddh Deshpande³, Felicity Park¹

¹Maternity and Gynaecology, John Hunter Hospital, the ²John Hunter Hospital, the ³Department of Paediatric Surgery, John Hunter Hospital

Background: Congenital volvulus is a rare, yet life-threatening condition that requires skillful diagnosis and management. Volvulus occurs when bowel loops become twisted around the mesenteric artery or its branches. The twisting of the mesenteric artery leads to congestion, impaired venous return and bowel necrosis.

Case description: We present a case of fetal ileal volvulus complicated by small bowel necrosis and meconium peritonitis. A third trimester ultrasound, prompted by decreased fetal movements and fetal tachycardia on CTG, showed fetal hydrops, an enlarged stomach, dilated bowel loops, ascites and a pericardial effusion. The peak systolic velocity in the middle cerebral artery was elevated, consistent with fetal anaemia. Daily surveillance ultrasound was performed and spontaneous labour occurred at 32 weeks' gestation. On day one of life, an abdominal x-ray detected pneumoperitoneum and an urgent laparotomy revealed an ileal volvulus with necrosis and meconium peritonitis. A segment of small bowel volvulus was resected and an end-to-end anastomosis performed.

Clinically, signs of congenital volvulus are non-specific, such as decreased fetal movements or increased fundal height, thus the diagnosis is often challenging. Ultrasound findings may include polyhydramnios, dilated bowel loops, gastric dilatation and ascites. Complications reported in the literature include perforation and haemorrhagic ascites, which may lead to anaemia, hypovolemia, heart failure and foetal demise.

Conclusion: This case highlights the importance of assessment of the fetal bowel in third trimester ultrasound, the complex diagnosis of fetal intestinal volvulus along with multidisciplinary team approach to management.

SOUND EXPOSURE IN INCUBATORS AND OPEN COTS IN A SINGLE-ROOM NEONATAL UNIT

Kobi Best^{1*}, Fiona Bogossian², Karen New²

¹University of Queensland & Gold Coast University Hospital, the ²University of Queensland

Background: In utero, the fetus is exposed to developmentally appropriate sounds which are attenuated by maternal uterine structures and amniotic fluid. The neonatal unit environment exposes neonates to suboptimal sensory stimulation, especially preterm neonates with immature neurosensory systems. The aim of this study was to measure sound exposure of both the internal and external environment of incubators and open cots.

Method: Sound meters were placed inside 51 occupied incubators and open cots, and within the surrounding environment within each single-room of the neonatal unit between September 15 and October 28, 2015. Data were recorded in decibels on an A-weighted scale (dBA), in one-second increments over a minimum of 48 hours per room.

Results: All average sound equivalent levels (Leq) were higher than the recommended 45 dBA for both incubators and open cots. The Leq inside the incubator was higher than the surrounding environment for day (75 dBA vs. 68.85 dBA) and night (74.91 dBA vs. 68.24 dBA). Incubators were observed to have a higher total percentage of time above 45 dBA than open cots for both day (99.84% vs. 75.59%) and night (98.66% vs. 59.56%). Maximum sound levels (Lmax) peaked at 126.7 dBA inside the incubator and 124.7 dBA inside the open cot.

Conclusion: Sound levels in the neonatal unit are all higher than recommendations. The long-term sequelae of adverse sensory exposure on preterm neurodevelopment remains unclear. Renewed attention on factors contributing to increased sound levels in the neonatal unit need to be addressed.

TYPES OF OBSERVATION CHARTS IN AUSTRALIAN AND NEW ZEALAND NICUs (TOOCAN)

Kim Bland^{1*}, Alison L Kent¹, Nicholas Sanders¹

¹Centenary Hospital for Women and Children

Introduction: Paper-based recording of vital signs is the cornerstone of monitoring and detecting deteriorating patients, especially so in critical care environments such as NICU. The evidence suggest that graphical representation of data is superior to numerical. Currently there is no standardised observation chart for Australian and New Zealand NICU.

Methods: An electronic survey was emailed to a database of Neonatologist and Clinical Nurse Managers across Australia and New Zealand. Deidentified data was collected and analysed using descriptive analysis.

Results: Responses were received from 23% of those invited, representing 90% NICU and Neonatal Emergency Transport Services from Australia and New Zealand. The majority (77%) were aware of evidence supporting graphical representation of data, however 47% of those units who responded did not use graphical data on their charts. Opinion regarding standardised chart was very much divided, with 41% finding this idea acceptable, 49% being unsure and 10% finding it unacceptable for their units.

Conclusion: The importance of the observation chart for monitoring, observing and also detecting deteriorating patients is well established. Graphical representation of data has been shown to improve utilisation of this clinical tool. A standardised chart may also assist for staff who move across units. Currently almost half of Australian and New Zealand NICU and NETS do not utilise graphical data and the idea of a standardised chart was